

Multi Coder

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Multi Coder

Product Overview

Multi Coder is an audio encoder program designed to accept audio for a selected input source and accept the WideOrbit Automation for Radio Now Playing XML data. The interface allows for multiple output destinations with customized metadata embedded directly into the audio stream.

The encoder has a configurable Control port allowing the start and stop of individual streams with the use of TCP messaging.

The application allows for the configuration of one to ten independent encoders each accepting unique audio and metadata inputs.

Application Disclaimer: This tool was written with the help of Artificial Intelligence.

Getting Started

Windows Implementation

(MORE TO COME)

Installing with Linux

(MORE TO COME)

Getting Started

With the service running, and the UI opened in a browser the application is ready to use. For the initial use select System Settings from the Top banner.

The system setting options will populate the second panel in the UI. Each link will open the respective setting in the next Panel. Once all the settings are set, click on Save in the first panel (far left side of the page).

System Config

System config opens a set of system settings. These settings affect how the application will operate. With these settings the number of encoders can be set, along with some specific settings for each configured encoder.



Multi Coder

Multi-Coder
v2.0

Admin SettingsManualSign Out

Available Streams

<u>Encoder 1</u>	<u>Encoder 2</u>	<u>Encoder 3</u>	<u>Encoder 4</u>
AAC: Stopped	AAC: Stopped	AAC: Stopped	AAC: Stopped
MP3: Stopped	MP3: Stopped	MP3: Stopped	MP3: Stopped
HLS: Stopped	HLS: Stopped	HLS: Stopped	HLS: Stopped
SRT: Stopped	SRT: Stopped	SRT: Stopped	SRT: Stopped

Admin Settings

Select the Admin Settings for the Status Bar. You will be prompted to log in. the default user name is “Admin”, and default password is “change-me”

Admin Login ×

Username:

Password:

Login

Application Settings

Application settings will directly affect the functionality of the application. The sub menu will populate in Panel 2.

UI Port:

This set the port the application UI port will be accessed from. The default port is 8050. A change to the port will require a restart of the application for the changes to take effect.

Log Level:

Set the level of detail the system logs will write.

Info (Default): will write basic encoder information and diagnostics.

- Debug: Writes extremely detailed logs used for application debugging
- Warning: Writes a sparse log, only containing warning messages.



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Log Rotation

Designates the number of logs the application will maintain. The default number of logs kept in a single rotation is 5.

Log Retention (days)

Designates how long logs will be kept before they are purged. By default the application will maintain logs for 14 days before purging them.

Max Encoder Count

Sets the number of encoders that can be configured within this application. The default count is 5. Acceptable entries are 1 through 10.

Change Password

Change the system password. Enter the new password and confirm the new password.

Administrative Settings

Application Settings

Network Interfaces

Encoder Configuration

Application Settings

UI Port:

Log Level:

Log Rotation:

Log Retention (days):

Max Encoder Count:

Change Password:

Confirm Password:

Save Settings

Network Interfaces

This will provide a list of network interfaces detected on the host system. Checking or unchecking the box next to the interface will determine if the interface will be listed during the encoder configuration. This section will also allow a user to designate a friendly name for the network interface.

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Administrative Settings

Application Settings

Network Interfaces

Encoder Configuration

Network Interfaces

Assign friendly names to detected interfaces. Check the boxes to include an interface in encoder Network dropdowns.

- ☐ {{B7E65A13-D049-4773-AAE9-C066A5C88211}}
Ethernet
- ☒ {{1ADD1A0A-1286-4C51-B176-D10D652FA8A5}}
Business Network
- ☒ {{E98F5E13-2381-4341-8DBE-D4919E1D0800}}
Audio Network
- ☐ {{590617FD-3255-48CE-BC0C-F840B581DF26}}
Wi-Fi
- ☐ {{1BBAC41B-D2A4-490C-94FC-7E34DAD7EF0F}}
Bluetooth Network Connection
- ☐ {{A41EEB1B-1BBC-48C4-AC39-26B49BE54139}}
vEthernet (Audio LAN)
- ☐ {{74923735-18C9-11EF-B68C-806E6F6E6963}}
Loopback Pseudo-Interface 1
- ☐ {{210FF537-01D6-4787-A1F1-3BD4C9DDCC12}}
vEthernet (Default Switch)

Save

Encoder Configuration

Selecting Encoder Configuration will populate panel 2 with a list of encoders available. In this section it is possible to add or remove available encoders from the interface. Selecting an encoder will populate panel with the configuration options for the selected encoder.

AAC Listen Link

Provide a specific link where the AAC stream can be listened to.

MP3 Listen Link

Provide a specific link where the MP3 stream can be listened to.

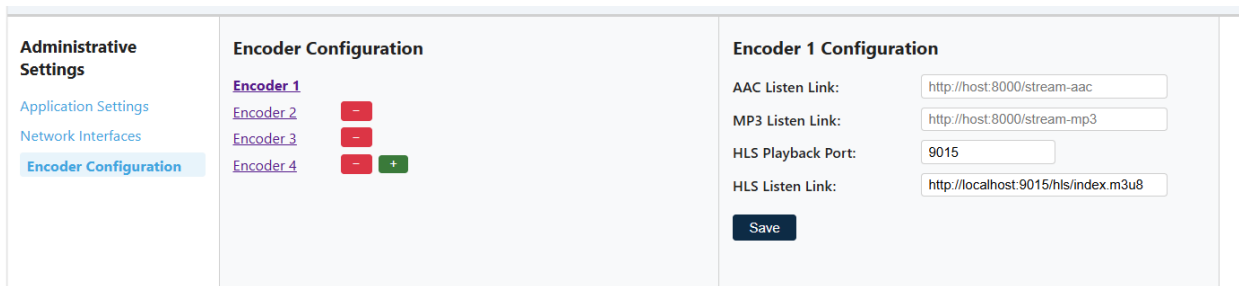
HLS Playback Port

Designates the port that will be used to broadcast the HLS audio stream.

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HLS Listen Link

Provide a specific link where the HLS stream can be listened to.



Administrative Settings

- Application Settings
- Network Interfaces
- Encoder Configuration**

Encoder Configuration

Encoder 1

Encoder 2

Encoder 3

Encoder 4

Encoder 1 Configuration

AAC Listen Link:

MP3 Listen Link:

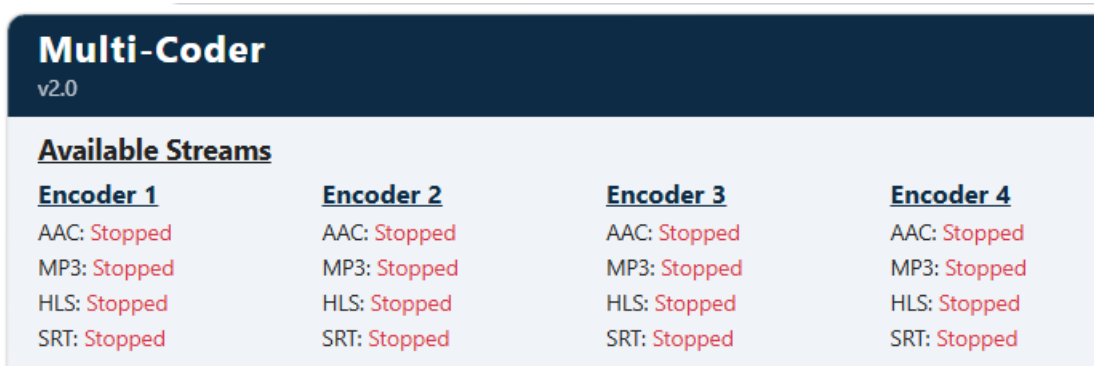
HLS Playback Port:

HLS Listen Link:

Save

Encoder Settings

Below the status banner a list of “Available Streams” will appear. Each encoder will display the status of the individual encoder streams. Click on the encoder name to display the encoder settings in the panels below.



Multi-Coder v2.0				
<u>Available Streams</u>				
<u>Encoder 1</u>	<u>Encoder 2</u>	<u>Encoder 3</u>	<u>Encoder 4</u>	
AAC: Stopped	AAC: Stopped	AAC: Stopped	AAC: Stopped	
MP3: Stopped	MP3: Stopped	MP3: Stopped	MP3: Stopped	
HLS: Stopped	HLS: Stopped	HLS: Stopped	HLS: Stopped	
SRT: Stopped	SRT: Stopped	SRT: Stopped	SRT: Stopped	

Configuring the Encoder

Input Selection

Input selection will set the input that will be used to generate the audio stream. In this section a user is presented with a VU meter to show the audio level of the audio on the selected input.

Input Gain

Adjusts the gain for the input audio.

Bitrate

Select the desired bit rate for the input audio

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Sample Rate

Sets the sample rate used for the input audio.

Input Source

Select the source for the input audio.

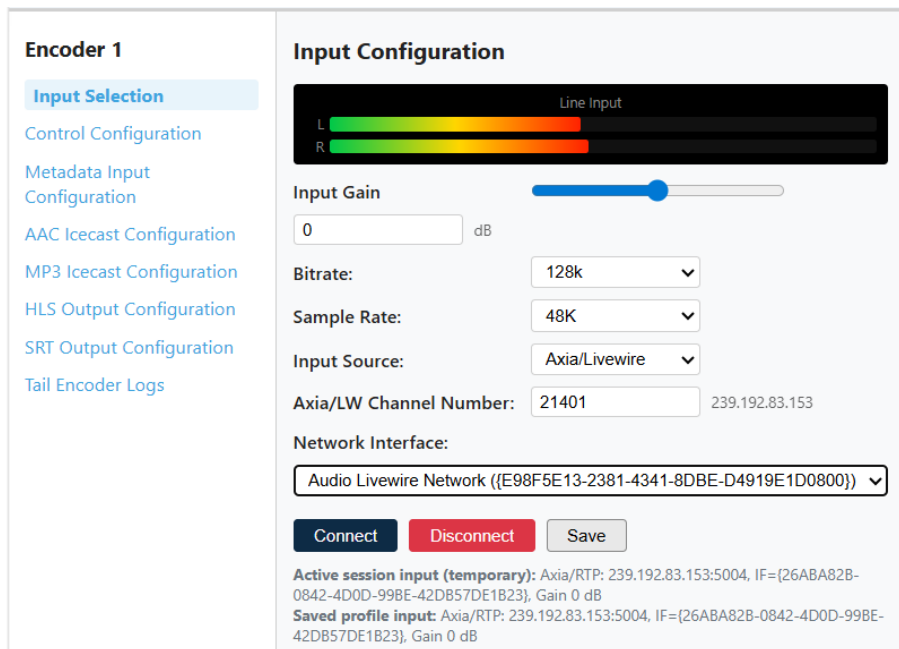
- **Axia / Livewire:** Uses an RTP stream. This setting allows the user to set the Axia audio channel. In the background the system will convert this channel number to a multicast address. This multicast address will be displayed next to the input channel number.
- **RTP:** present a dialog box to set the desired multicast address and port number that will be used for the input audio stream.
- **Audio Input:** Presents a dropdown menu of physical inputs detected on the host workstation.
- **SRT (Coming Soon):** Provides a list of settings to accept an Incoming SRT stream. (MORE DETAILS TO COME)

Network Interface

Sets the network interface that will be used for the network connection.

Connect / Disconnect / Save

Will Connect or Disconnect to the input audio. Save will save the settings to the Encoder configuration file.



The screenshot shows the 'Encoder 1' configuration window. On the left is a sidebar with a menu: 'Input Selection' (highlighted), 'Control Configuration', 'Metadata Input Configuration', 'AAC Icecast Configuration', 'MP3 Icecast Configuration', 'HLS Output Configuration', 'SRT Output Configuration', and 'Tail Encoder Logs'. The main area is titled 'Input Configuration'. It features a 'Line Input' level meter with L and R channels. Below the meter is an 'Input Gain' slider set to 0 dB. Further down are dropdown menus for 'Bitrate' (128k), 'Sample Rate' (48K), and 'Input Source' (Axia/Livewire). Below these is a text field for 'Axia/LW Channel Number' (21401) with the corresponding multicast address '239.192.83.153' displayed to its right. A 'Network Interface' dropdown is set to 'Audio Livewire Network ({E98F5E13-2381-4341-8DBE-D4919E1D0800})'. At the bottom are three buttons: 'Connect' (dark blue), 'Disconnect' (red), and 'Save' (grey). A status section at the very bottom shows 'Active session input (temporary): Axia/RTP: 239.192.83.153:5004, IF={26ABA82B-0842-4D0D-998E-42DB57DE1B23}, Gain 0 dB' and 'Saved profile input: Axia/RTP: 239.192.83.153:5004, IF={26ABA82B-0842-4D0D-998E-42DB57DE1B23}, Gain 0 dB'.

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Control Configuration

Multi Coder provides the ability to start and stop individual streams with TCP messages.

Control Port

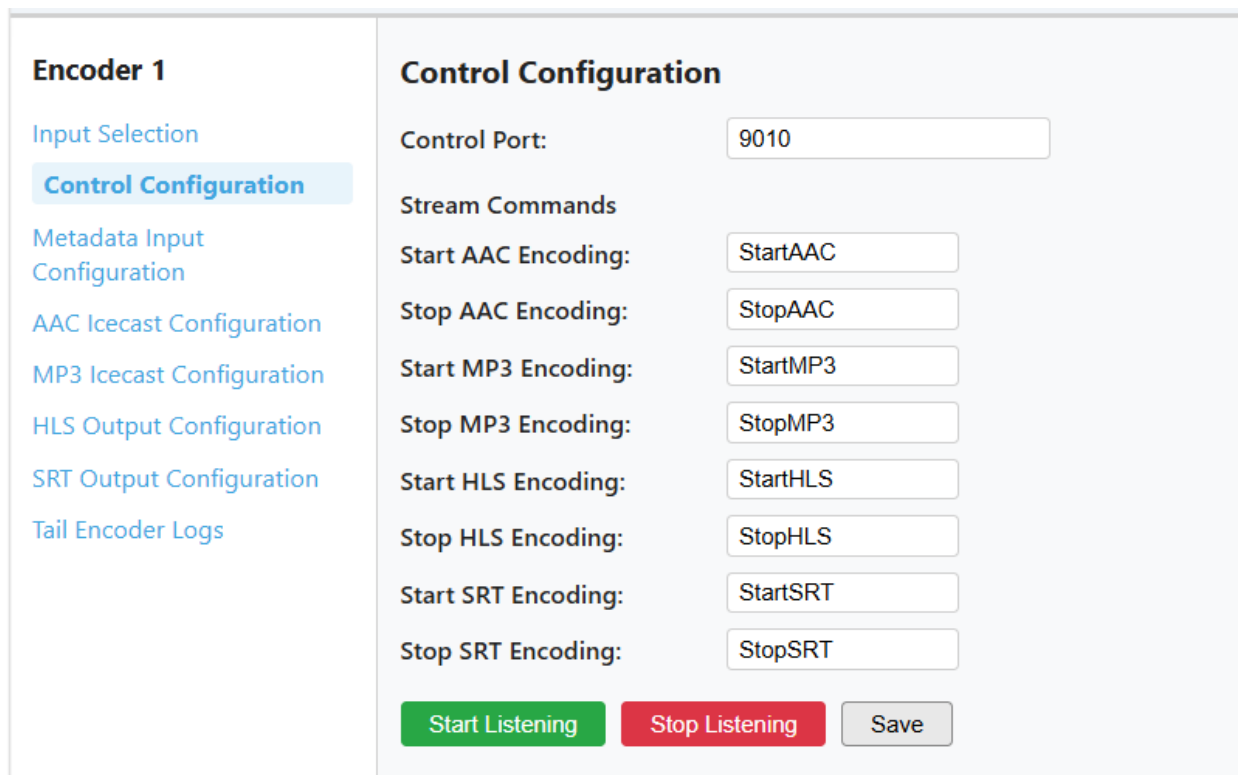
Designate the listening port for the incoming TCP control messages.

Stream Commands

Set the specific message to listen for to control the individual streams. The default message string is listed for each command. It is possible for the user to define the specific messages that will be listened for.

Start / Stop Listening

Start Listening will open the designated port and watch for the incoming messages. Stop Listening will close the port so no messages are received.



The screenshot shows the 'Encoder 1' configuration page. On the left is a sidebar with links: 'Input Selection', 'Control Configuration' (highlighted), 'Metadata Input Configuration', 'AAC Icecast Configuration', 'MP3 Icecast Configuration', 'HLS Output Configuration', 'SRT Output Configuration', and 'Tail Encoder Logs'. The main area is titled 'Control Configuration'. It contains a 'Control Port' field with the value '9010'. Below this is a section for 'Stream Commands' with pairs of labels and input fields: 'Start AAC Encoding:' / 'StartAAC', 'Stop AAC Encoding:' / 'StopAAC', 'Start MP3 Encoding:' / 'StartMP3', 'Stop MP3 Encoding:' / 'StopMP3', 'Start HLS Encoding:' / 'StartHLS', 'Stop HLS Encoding:' / 'StopHLS', 'Start SRT Encoding:' / 'StartSRT', and 'Stop SRT Encoding:' / 'StopSRT'. At the bottom are three buttons: 'Start Listening' (green), 'Stop Listening' (red), and 'Save' (grey).

Metadata Input Configuration

Multi Coder can accept WO Automation for Radio XML now playing data. Once the data has been received, settings for each individual stream will parse the data and embed the data directly into the stream.

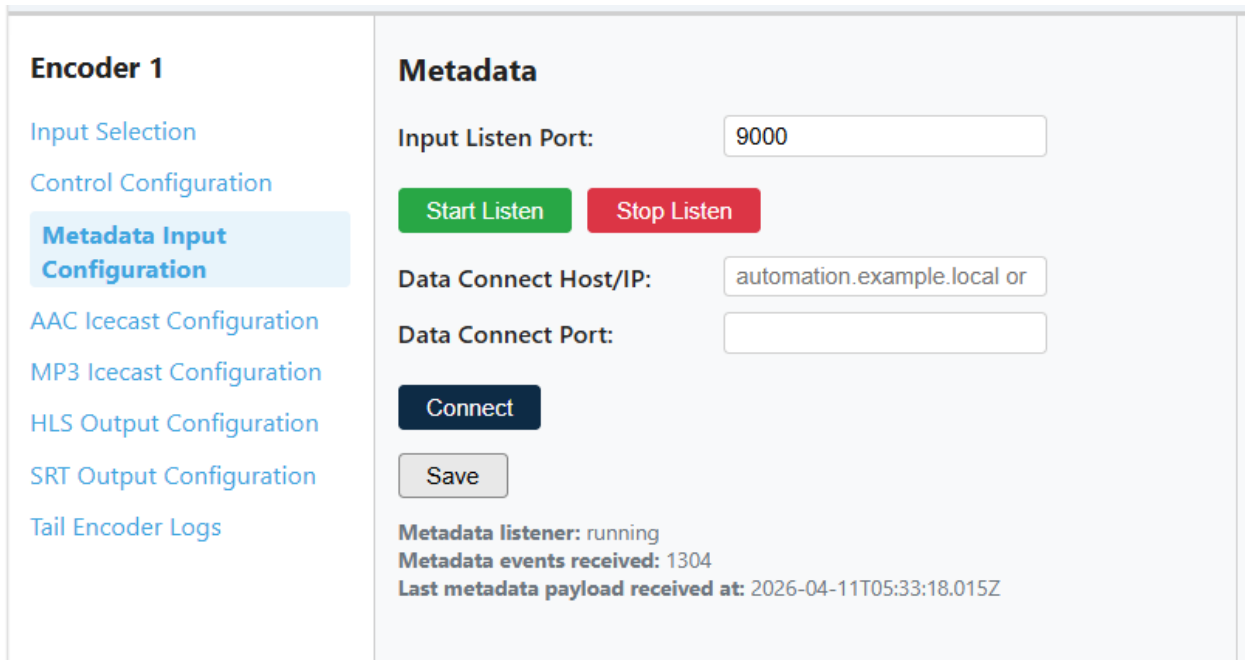
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Input Listen Port:

Sets the port to open to accept incoming XML data from the automation system. Pressing Start or Stop Listening will open or close the designated port. In this configuration the encoder would act as the “server”

Data Connect

Set the IP address and Port number to have the encoder connect directly to the automation system. In this configuration the encoder is acting as the “client” to the automation system. Pressing Connect, the encoder connect to the automation system based on the set IP and port number.



The screenshot displays the 'Encoder 1' configuration page. On the left, a sidebar lists navigation options: 'Input Selection', 'Control Configuration', 'Metadata Input Configuration' (highlighted), 'AAC Icecast Configuration', 'MP3 Icecast Configuration', 'HLS Output Configuration', 'SRT Output Configuration', and 'Tail Encoder Logs'. The main content area is titled 'Metadata' and contains the following elements:

- Input Listen Port:** A text input field containing the value '9000'.
- Start Listen** (green button) and **Stop Listen** (red button) buttons.
- Data Connect Host/IP:** A text input field containing the placeholder text 'automation.example.local or'.
- Data Connect Port:** An empty text input field.
- Connect** (dark blue button) and **Save** (light gray button) buttons.
- Status text at the bottom: 'Metadata listener: running', 'Metadata events received: 1304', and 'Last metadata payload received at: 2026-04-11T05:33:18.015Z'.

AAC and MP3 Icecast Configuration

The settings are the same for each of these streams.

Icecast URL

Sets the url the encoder will connect to deliver the audio stream

User Name

Sets the user name for the stream to connect to the Streaming Provider

Password

Sets the Password required for the streaming provider

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IcyMetaInt Port

If the provider requires a specific port for data. Default port is 8192.

Station ID

Sets the Station ID if it is required by the streaming provider.

Send Embedded Metadata

A dropdown with YES or NO options. Selecting YES will embed the audio stream with the metadata. NO will not send metadata

Network Interface

Designates the network interface that will be used to send the stream.

Edit Metadata Parser

Opens the configuration for the metadata parser in panel 3.

Start / Stop / Save

Starts or Stops the selected Encoder. Save will save the settings for the encoder.

Encoder 1
[Input Selection](#)
[Control Configuration](#)
[Metadata Input Configuration](#)
[AAC Icecast Configuration](#)
[MP3 Icecast Configuration](#)
[HLS Output Configuration](#)
[SRT Output Configuration](#)
[Tail Encoder Logs](#)

AAC Icecast Configuration

Icecast AAC URL:

User Name:

Password:

IcyMetaInt Port:

Station ID:

Send Embedded Metadata: ▼

Network Interface:
 ▼

[Edit Metadata Parser](#)

Listen Link:
Not configured — set in Admin > Encoder Configuration

Start Encoder

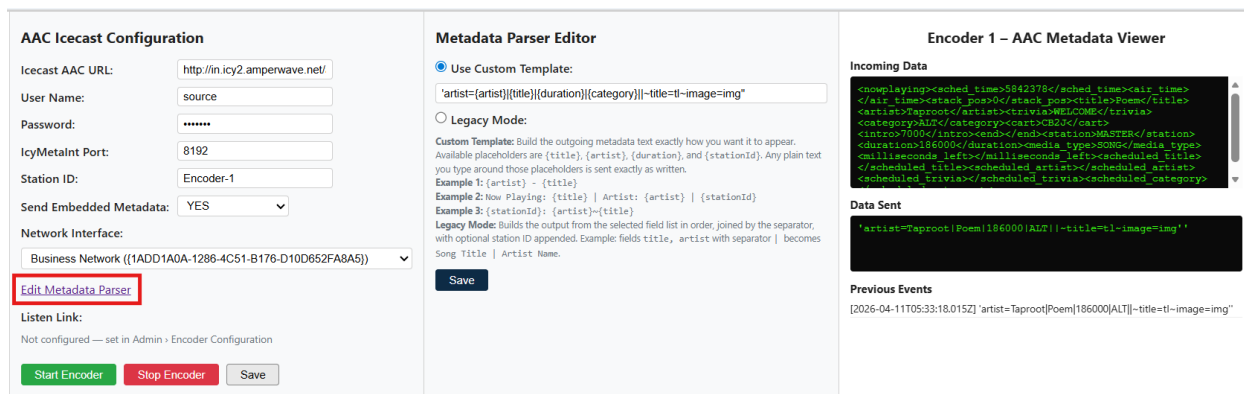
Stop Encoder

Save

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Icecast metadata Parser

When selected as part of the AAC or MP3 encoder, a configuration screen will open in panel 3. Panel 4 will provide a display of the incoming metadata, and a result of the parsed data. A log of previous items is provided at the bottom of panel 4.

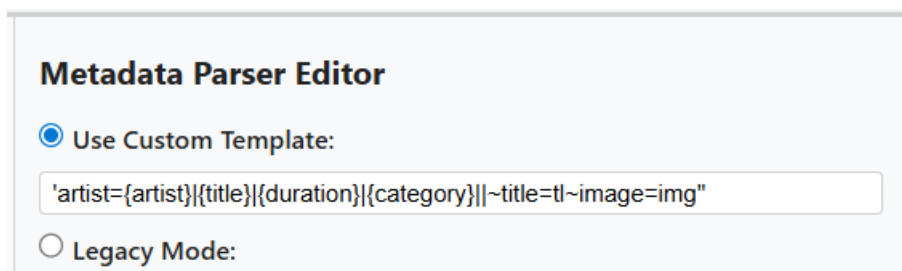


The screenshot displays three panels from the MultiCoder application:

- AAC Icecast Configuration:** Contains fields for Icecast AAC URL, User Name, Password, IcyMetalint Port, Station ID, Send Embedded Metadata (set to YES), and Network Interface. A red box highlights the "Edit Metadata Parser" link.
- Metadata Parser Editor:** Shows the "Use Custom Template" option selected. The custom template is: `'artist={artist}|{title}|{duration}|{category}||~title=tl-image=img'`. It also includes a "Legacy Mode" option and a "Save" button.
- Encoder 1 - AAC Metadata Viewer:** Displays "Incoming Data" as a large block of XML metadata and "Data Sent" as a single line of parsed metadata: `'artist=Taproot|Poem|186000|ALT||~title=tl-image=img'`. It also shows "Previous Events" with a timestamped log entry.

Use Custom Template

Allows for a customized data stream to be set. Using place holders containing the name of the specific tags from the XML feed. Place holders will be defined by placing the tag name inside {} brackets. Example: {artist} will display the Artist filed from the XML feed.



This close-up shows the "Metadata Parser Editor" with the "Use Custom Template" radio button selected. The text input field contains the custom template: `'artist={artist}|{title}|{duration}|{category}||~title=tl-image=img'`.

Legacy Mode

With Legacy Mode selected, a user is able to define a separator, and the fields to include in the data feed. Example: set the separator of | and include the fields title, artist results in title | artist.

Include Station ID will determines if the station ID as defined on the encoder settings will be included with the data.

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Metadata Parser Editor

☐ Use Custom Template:

☒ Legacy Mode:

Separator:

Include Station ID: ☒

Include Fields:

Custom Template: Build the outgoing metadata text exactly how you want it to appear. Available placeholders are {title}, {artist}, {duration}, and {stationId}. Any plain text you type around those placeholders is sent exactly as written.
Example 1: {artist} - {title}
Example 2: Now Playing: {title} | Artist: {artist} | {stationId}
Example 3: {stationId}: {artist}~{title}

Legacy Mode: Builds the output from the selected field list in order, joined by the separator, with optional station ID appended. Example: fields title, artist with separator | becomes Song Title | Artist Name.

HLS Output Configuration

Segment Length

Sets the length each HLS audio segment will be. Default Setting is 5 seconds.

Buffer Window

Sets the number of buffered segments that will be maintained.

Data offset

Allows the embedded data to be offset to accommodate any delays that may be set. Best practice is to start at 0 and adjust down if needed.

Low Latency

Checking this box will shift the HLS feed to a Low Latency setting.

Embed Metadata

Determines if metadata will be embedded in the HLS stream

Network Interface

Designates the network interface the HLS stream will publish on.

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Admin and Live Stream links

- Admin Listen Link: this is set as part of administrative settings for the encoder. This is a custom link the administrator can set.
- Live Listen Link: will automatically display the link generated by the encoder based on the network interface and port designated.

Encoder 1
[Input Selection](#)
[Control Configuration](#)
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[MP3 Icecast Configuration](#)
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[Tail Encoder Logs](#)

HLS Configuration

Segment Length (seconds):

Buffer Window (segments):

Data Offset Time (seconds):

Low Latency Stream: ☐

Embed Metadata: ☒

Network Interface:

Business Network ({1ADD1A0A-1286-4C51-B176-D10D652FA8A5})

[Edit Metadata Parser](#)

Admin Listen Link: <http://localhost:9015/hls/index.m3u8>

Live Stream Link: <http://localhost:9015/hls/index.m3u8>

Start HLSStop HLSSave

Metadata Parser

Opens up a configuration page in panel 3 to set how the metadata will be parsed for this stream.

Parsing Method

Determines the how the data will be embedded into the HLS stream.

- EXT Tags: will set the parsed XML data into the EXTIF tags of the HLS stream
- ID3: Will set the parsed XML data to the ID3 tags of the HLS stream
- XML Pass Through: simple passed the XML data to the HLS stream

Embedded Scope

- Current: only sends the data for the current media playing.
- Current + Future: If provided by the automation system, the encoder will pass the current media playing, plus the next 5 elements set to play.

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XML Tags to Use:

Select the tags from the XML feed that will be used as part of the embedded feed. The most common tags are expressed with check boxes. For any custom fields enter in the tag name separated by a comma.

Metadata Parser Editor

Parsing Method:

Embed Scope:

XML Tags To Use:

☒ sched_time	☒ stack_position
☒ title	☒ artist
☒ trivia	☒ category
☒ cart	☒ media_type
☐ station	☐ stationId (from stream config)

Custom Tags:

XML Pass Through: emits raw XML. ID3: emits ID3-like frames (TIT2/TPE1/etc). EXT: emits compact JSON for EXT-style playlist metadata. Scope controls whether only current item or current+future stack is emitted. stationId is read from the stream config, not from the XML payload.

Panel 4 will provide a display of the incoming XML data, and the parsed data going out.

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Encoder 1 – HLS Metadata Viewer

Incoming Data

```
<nowplaying><sched time>5842378</sched time><air time>
</air time><stack_pos>0</stack_pos><title>Poem</title>
<artist>Taproot</artist><trivia>WELCOME</trivia>
<category>ALT</category><cart>CB2J</cart>
<intro>7000</intro><end></end><station>MASTER</station>
<duration>186000</duration><media_type>SONG</media_type>
<milliseconds_left></milliseconds_left><scheduled_title>
</scheduled_title><scheduled_artist></scheduled_artist>
<scheduled_trivia></scheduled_trivia><scheduled_category>
```

Data Sent

```
{"current":
{"type":"SONG","title":"Poem","album":"WELCOME","artist":"Tap
[{"type":"UNSPECIFIED","title":"CLASSIC SNACK I.D.
#01","album":"THIS IS...","artist":"THIS
IS...","image":"","duration":6.100,"start":1775885784.018,"id
{"type":"SONG","title":"The Day I Tried To
Live","album":"SUPERUNKNOWN","artist":"Soundgarden","image":
{"type":"UNSPECIFIED","title":"CRANK IT UP LOUD WKND I.D.
```

Previous Events

```
[2026-04-11T05:33:18.015Z] {"current":
{"type":"SONG","title":"Poem","album":"WELCOME","artist":"Taproot","image":"","duration":
[{"type":"UNSPECIFIED","title":"CLASSIC SNACK I.D. #01","album":"THIS
IS...","artist":"THIS
IS...","image":"","duration":6.100,"start":1775885784.018,"id":"0073","category":"IMG"},
{"type":"SONG","title":"The Day I Tried To
Live","album":"SUPERUNKNOWN","artist":"Soundgarden","image":"","duration":314.000,"st
{"type":"UNSPECIFIED","title":"CRANK IT UP LOUD WKND I.D. #02","album":"NO
BONUS CODE","artist":"NO BONUS
CODE","image":"","duration":6.400,"start":1775886104.118,"id":"0097","category":"IMG"},
{"type":"SPOT","title":"Jelli 150","album":"jelli_150","artist":"C-AGY GOOD CAR
KEEPING","image":"","duration":15.100,"start":1775886110.518,"id":"COX1","category":"CO
```

SRT Output Configuration

(COMING SOON)

Multi Coder

Tail Encoder Logs

Will display the diagnostic log for the selected encoder

Encoder 1

Input Selection

Control Configuration

Metadata Input
Configuration

AAC Icecast Configuration

MP3 Icecast Configuration

HLS Output Configuration

SRT Output Configuration

Tail Encoder Logs

```
wledged by worker
[2026-04-11T05:34:11.841Z] [Encoder-1] Heartbeat -
AAC:stopped MP3:LIVE HLS:LIVE SRT:stopped
[2026-04-11T05:34:12Z] [Supervisor] MP3 stop reque
sted
[2026-04-11T05:34:12.966Z] [Encoder-1] Control com
mand: StopMP3
[2026-04-11T05:34:12.972Z] [Encoder-1] MP3 Icecast
stopped
[2026-04-11T05:34:12Z] [Supervisor] MP3 stop ackno
wledged by worker
[2026-04-11T05:34:13.068Z] [Encoder-1] Metadata cl
ient connected but no payload yet: 192.168.50.214:
65346
[2026-04-11T05:34:28.094Z] [Encoder-1] Metadata cl
ient connected but no payload yet: 192.168.50.214:
65346
[2026-04-11T05:34:42.117Z] [Encoder-1] Heartbeat -
AAC:stopped MP3:stopped HLS:LIVE SRT:stopped
[2026-04-11T05:34:43.102Z] [Encoder-1] Metadata cl
ient connected but no payload yet: 192.168.50.214:
65346
[2026-04-11T05:34:58.148Z] [Encoder-1] Metadata cl
ient connected but no payload yet: 192.168.50.214:
65346
[2026-04-11T05:35:12.468Z] [Encoder-1] Heartbeat -
AAC:stopped MP3:stopped HLS:LIVE SRT:stopped
[2026-04-11T05:35:13.153Z] [Encoder-1] Metadata cl
ient connected but no payload yet: 192.168.50.214:
65346
[2026-04-11T05:35:28.177Z] [Encoder-1] Metadata cl
ient connected but no payload yet: 192.168.50.214:
65346
[2026-04-11T05:35:42.757Z] [Encoder-1] Heartbeat -
AAC:stopped MP3:stopped HLS:LIVE SRT:stopped
[2026-04-11T05:35:43.205Z] [Encoder-1] Metadata cl
ient connected but no payload yet: 192.168.50.214:
65346
[2026-04-11T05:35:58.226Z] [Encoder-1] Metadata cl
ient connected but no payload yet: 192.168.50.214:
65346
[2026-04-11T05:36:12.966Z] [Encoder-1] Heartbeat -
AAC:stopped MP3:stopped HLS:LIVE SRT:stopped
[2026-04-11T05:36:13.232Z] [Encoder-1] Metadata cl
ient connected but no payload yet: 192.168.50.214:
65346
```